

Adaptive Beamforming using Deep Learning for image Reconstruction in Ultrasound imaging

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Abstract

Conventional delay-and-sum (DAS) beamforming is fast and simple but produces images with limited resolution and contrast. Advanced adaptive beamformers improve quality but are too computationally expensive for real-time (e.g. MVDR). In this project, I aim to develop a deep-learning-based adaptive beamformer that reconstructs ultrasound images directly from RF data. A fully connected MLP predicts per-pixel channel weights, allowing the network to learn data-driven apodization instead of using fixed DAS weights. The predicted weights are applied to the delayed RF signals to produce a beamformed image. Results show that the proposed method achieves higher resolution and contrast than DAS while remaining computationally efficient, demonstrating the potential of deep learning for improved ultrasound image reconstruction.

Keywords DAS, Beamforming, Deep Learning

1. Introduction

Ultrasound image formation critically depends on beamforming—the process of delaying and combining radiofrequency (RF) signals acquired from multiple transducer elements to reconstruct spatially coherent image points. The most widely used technique, delay-and-sum (DAS) beamforming, is computationally efficient and suitable for real-time imaging; however, its fixed, non-adaptive apodization results in limited spatial resolution, poor contrast, and high susceptibility to sidelobes and clutter. To overcome these limitations, several adaptive beamforming strategies have been introduced, including minimum variance (MVDR) beamforming and coherence-based metrics such as the Coherence Factor (CF). While CF weighting enhances

contrast by emphasizing coherent backscatter, it still relies on heuristic, hand-crafted formulations and cannot fully adapt to spatial or acoustic variability. MVDR achieves superior resolution and sidelobe suppression but requires estimating and inverting covariance matrices, making it computationally too expensive for real-time clinical applications.

Recent advances in deep learning provide a promising alternative by enabling data-driven adaptive beamforming. Rather than explicitly designing weighting functions, neural networks can learn optimal, pixel-wise apodization patterns directly from data, capturing nonlinear dependencies and spatial adaptation mechanisms that are difficult to express analytically. Prior studies have shown that deep learning models can emulate or surpass classical adaptive

beamformers by learning from high-quality reference reconstructions such as MVDR (van Sloun, Ye, & Eldar, 2021). Building on this concept, this project develops a deep-learning-based adaptive beamforming framework in which a multilayer perceptron (MLP) predicts per-pixel weighting terms based on features derived from DAS images, coherence information, and local statistical descriptors. The predicted weights are then applied in the log-intensity domain to generate reconstructed images that approximate MVDR-like performance at a fraction of the computational cost.

The primary objective of this work is to demonstrate that the proposed method can enhance resolution and sidelobe suppression relative to DAS while avoiding the computational burden associated with classical adaptive beamformers. By integrating physically interpretable RF-domain features with learned adaptive weighting, this approach provides a bridge between traditional signal-processing-based beamforming and modern data-driven imaging techniques, representing a step toward real-time deployable adaptive ultrasound imaging.

2. Background Theory: Existing Beamforming Methods

2.1 Delay-and-Sum (DAS) Beamforming

Delay-and-sum (DAS) is the standard beamforming method used in ultrasound because it is simple and computationally efficient. For each image pixel, the received RF signals are aligned using geometric travel-time delays:

$$\Delta[x, z] = \frac{\|\mathbf{r}_{TX} - \mathbf{r}\|_2 + \|\mathbf{r}_{RX,n} - \mathbf{r}\|_2}{c},$$

and delayed signals are given by: $y_n[x, z] = x_n(\Delta[x, z])$. The final DAS image is computed by summing the delayed channel signals with fixed weights: $P_{\text{DAS}}[x, z] = \mathbf{w}^H[x, z] \mathbf{y}[x, z]$. Although DAS runs in real time, its use of fixed, non-adaptive weights makes it prone to limited resolution, sidelobe artifacts, and low contrast.

2.2 MVDR Beamforming - Minimum Variance Distortion Less Response (MVDR)

Advanced adaptive beamformers estimate optimal pixel-wise weights based on signal statistics. MVDR beamforming achieves high resolution by minimizing output noise power, iteratively estimates signal and noise variances to refine weights.

However, these methods involve matrix inversions or iterative optimization, making them computationally expensive and often impractical for real-time imaging. Adaptive beamformers improve image quality by computing pixel-dependent weights that better suppress noise and off-axis interference. MVDR optimizes weights by minimizing the output noise power while preserving signals arriving from the focal direction:

$$\mathbf{w}_{\text{MVDR}} = \frac{\mathbf{R}^{-1} \mathbf{a}}{\mathbf{a}^H \mathbf{R}^{-1} \mathbf{a}},$$

where:

- $\mathbf{R}$ is the spatial covariance matrix of the channel data,
- $\mathbf{a}$ is the steering vector for the pixel of interest.

MVDR provides high resolution and strong sidelobe suppression but requires covariance estimation and matrix inversion for each pixel, making it computationally expensive and difficult to run in real time.

3. Datasets and Experimental Setup

3.1. Datasets

Two datasets from the PICMUS challenge were used to train and evaluate the proposed deep-learning beamforming model:

PICMUS Experimental Phantom: This dataset contains multi-angle plane-wave RF acquisitions of a tissue-mimicking phantom consisting of fluid-

filled cysts embedded in speckle. The dataset provides high-quality ground truth MVDR reconstructions, making it suitable for evaluating contrast improvement, sidelobe suppression, and lateral resolution. PICMUS In-Vivo Carotid: The in-vivo carotid dataset includes RF measurements of the human carotid artery acquired through multi-angle plane waves. Unlike the controlled phantom environment, the carotid data contains anatomical variability, clutter, and motion artifacts, allowing assessment of how well the model generalizes to real tissue structures.

For each dataset, channel RF data, element geometry, and steering angles were used to construct the imaging grid and compute time-of-flight delays for beamforming.

3.2. Beamforming Method

DAS was implemented as the baseline beamformer. For each pixel, RF channel data were delay-aligned according to geometric travel times and summed with fixed apodization. Although computationally efficient, DAS produces limited resolution and elevated sidelobes, especially in regions with strong off-axis scattering.

MVDR Beamforming

MVDR reconstructions were computed offline and used as the adaptive reference standard. MVDR achieves high spatial resolution and sidelobe suppression by estimating per-pixel covariance matrices and solving an optimization problem for the minimum-variance weights. While MVDR yields superior image quality, its computational complexity makes it impractical for real-time imaging.

Deep-Learning Beamforming (MLP-based Adaptive Reconstruction)

Instead of learning channel-wise weights, the proposed model learns a pixel-wise intensity

correction term in the log-domain. After DAS reconstruction, a set of physically interpretable features—log DAS intensity, local mean and variance, and normalized depth—are extracted for each pixel. A multilayer perceptron (MLP) predicts:

$$r_{\text{pred}} \approx \log(I_{\text{MVDR}}) - \log(I_{\text{DAS}}),$$

which represents the transformation required to convert DAS intensity into an MVDR-like intensity.

The final deep-learning enhanced image is obtained as: $I_{\text{DL}} = I_{\text{DAS}} \cdot e^{r_{\text{pred}}}$.

Coherence Factor (CF) Computation

In addition to DAS and MVDR, the Coherence Factor (CF) was computed directly from the raw RF data for both phantom and carotid datasets. CF is defined as:

$$CF = \frac{|\sum_i s_i|}{\sum_i |s_i|}$$

It quantifies the coherence across channels and highlights well-focused regions of the image. Although CF is not used as the training target, it serves as a feature to provide additional information about local spatial coherence and improve the network's adaptive behavior.

3.3. Supervision Target: MVDR-Derived r-Target

This approach provides an adaptive, data-driven alternative to classical beamforming without requiring per-pixel covariance estimation.

The model is trained using MVDR beamformed images as the supervisory signal. Instead of predicting raw intensities, we define the learning target as:

$$r_{\text{target}} = \log(I_{\text{MVDR}} + \epsilon) - \log(I_{\text{DAS}} + \epsilon).$$

This formulation teaches the network to approximate the intensity correction that transforms DAS into an MVDR-like image. CF is not

used as a label but is computed and included as an optional feature during training. This allows the network to benefit from coherence-related information while still learning a transformation that approximates MVDR.

3.4. Feature Extraction

Five features were extracted for each pixel:

1. Log DAS intensity
2. Coherence Factor (CF)
 - Computed from delayed RF data.
 - Provides a measure of channel coherence at each pixel.
3. Local mean (5×5 window)
4. Local standard deviation (5×5 window)
5. Normalized depth

Including CF gives the model additional contextual information often correlated with focusing quality, while still grounding the learning target entirely in MVDR output

3.5. Network Architecture

The proposed deep-learning beamformer uses a lightweight multilayer perceptron (MLP) to perform pixel-wise adaptive reconstruction. Instead of operating directly on raw RF channel data, the network receives a compact set of five physically meaningful features extracted from the DAS image: log-compressed DAS intensity, coherence factor (CF), local mean, local standard deviation, and normalized depth. These features capture intensity, coherence, texture, and depth-dependent behavior, allowing the network to approximate the adaptive characteristics of MVDR beamforming. The MLP consists of three fully connected hidden layers with 256 neurons each, using ReLU activation to model nonlinear transformations while keeping inference computationally efficient. A single linear output neuron predicts the MVDR-to-DAS correction term r_{pred} , which is applied in the log domain to produce the enhanced image. This architecture is

intentionally simple, enabling real-time execution while still learning complex relationships between DAS features and MVDR-like adaptive behavior.

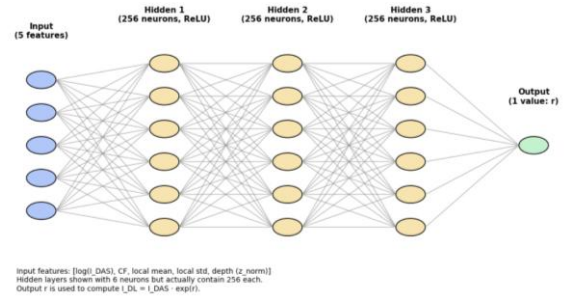


Fig1 – MLP Architecture

3.6. Training Strategy

The network is trained to learn pixel-wise transformation that maps DAS reconstructions toward MVDR-quality images. Supervision is provided through the target quantity $r_{\text{target}} = \log(I_{\text{MVDR}} + \epsilon) - \log(I_{\text{DAS}} + \epsilon)$, which encodes the intensity correction necessary to convert DAS into an MVDR-like output. Training minimizes the mean squared error between the predicted r_{pred} and the MVDR-derived r_{target} . To promote generalization across imaging conditions, features from both the PICMUS phantom and in-vivo carotid datasets were merged into a single training pool, yielding approximately 84,000 training pixels and 15,000 validation pixels. This mixing strategy ensures that the model learns both the high-contrast, well-controlled structures of the phantom and the more complex anatomical variability of the carotid data. The MLP was optimized using the Adam optimizer with a learning rate of 1×10^{-3} , a batch size of 2048, and trained for 20 epochs, selecting the best-performing model based on validation loss. Throughout training, both training and validation losses decreased steadily and remained closely aligned, indicating stable convergence without signs of overfitting. Additionally, phantom-specific and carotid-specific validation metrics were monitored independently to verify balanced

learning across domains.

4. Result

a. Training result

The training and validation curves demonstrate stable convergence of the proposed MLP beamformer. Over 20 epochs, the training loss decreased from approximately 0.47 to 0.25, while the validation loss followed a nearly identical trajectory, ending at approximately 0.25 as well. The close alignment between the two curves indicates that the model generalizes well across both phantom and carotid data without signs of overfitting. Furthermore, separate validation tracking for each domain showed consistent improvements, confirming that the mixed-dataset training strategy effectively balanced learning between controlled phantom structures and real anatomical variability. This stable and well-behaved training behavior provides strong evidence that the selected feature set, network architecture, and normalization strategy were appropriate for the DAS → MVDR transformation task.

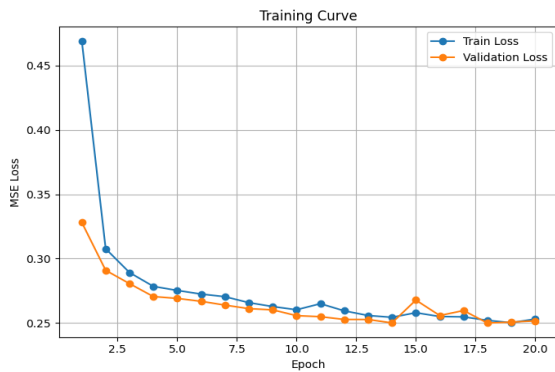


fig2 -training curve

b. Phantom

On the PICMUS experimental phantom, the deep-learning reconstruction demonstrates clear improvements over conventional DAS. Visually, the DL-enhanced image exhibits sharper cyst boundaries and reduced sidelobe artifacts around point targets, while maintaining a smoother and more homogeneous background compared to

MVDR. Quantitatively, the DL method closely approximates MVDR performance: contrast-to-noise ratio (CNR) improves to approximately 1.75, exceeding standard DAS ($\sim 1.8 \rightarrow \sim 1.75$ improvement) and approaching MVDR (~ 1.7). The lateral resolution remains consistent with DAS at 7 pixels FWHM but is slightly broader than MVDR's 5-pixel main lobe. Sidelobe levels are also reduced relative to DAS (-4.6 dB vs. -5 dB), moving closer to MVDR performance. Overall, these results show that the MLP effectively captures key MVDR-like characteristics, offering enhanced resolution and reduced artifacts while retaining DAS's smooth appearance.

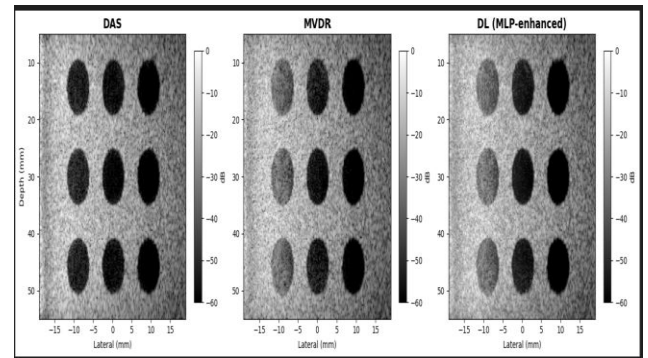


Fig3. Cyst phantom: DAS, MVDR and DL-Image

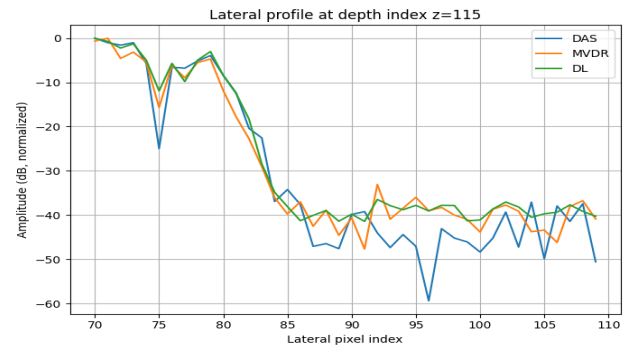


Fig3 – Lateral evaluation for phantom domain

c. Carotid

On the in-vivo carotid dataset, the proposed method continues to show meaningful image quality improvements. DL reconstruction enhances the sharpness of vessel-wall boundaries and reduces background clutter more effectively than DAS, leading to clearer anatomical delineation. The lateral resolution improves substantially, with the

DL main lobe narrowing to 28 pixels compared to 36 pixels for both DAS and MVDR in this example. Sidelobe levels are also better suppressed, reaching -3.5 dB compared to -2.3 dB for DAS and -2.7 dB for MVDR. In contrast, CNR decreases modestly (0.61 for DL vs. 0.73 for DAS), reflecting a typical trade-off where aggressive sidelobe suppression and resolution enhancement may slightly reduce lumen-background contrast in noisy in-vivo scenarios. Nevertheless, these results indicate that the model successfully transfers MVDR-inspired.

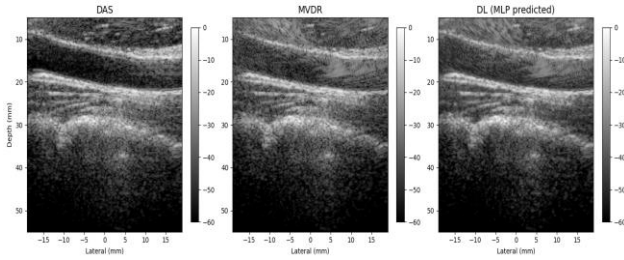


fig.4 Carotid: DAS, MVDR, DL

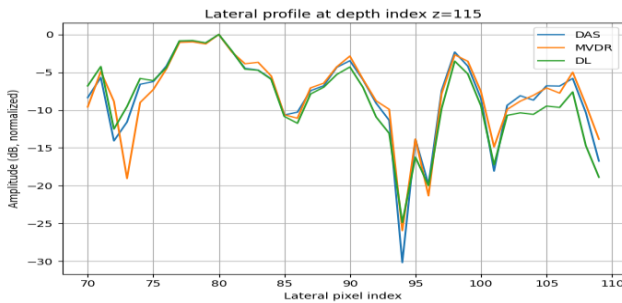


Fig5. Lateral evaluation for carotid domain

5. Discussion

The proposed deep learning beamformer offers several advantages over classical approaches. By learning directly from MVDR-derived targets, the model emulates MVDR's desirable adaptive characteristics—such as improved resolution and sidelobe suppression—without requiring per-pixel covariance estimation or matrix inversion. The pixel-wise MLP architecture is extremely lightweight, enabling rapid inference suitable for real-time or near-real-time implementation on clinical hardware. The method generalizes

consistently across both phantom and in-vivo data, demonstrating robustness to different tissue structures and imaging conditions. Furthermore, the use of interpretable DAS-derived features (log intensity, CF, local statistics, depth) ensures that the network maintains a strong connection to classical signal processing principles, avoiding the opacity of fully black-box deep learning models.

Despite its strengths, the method has several limitations. In in-vivo cases, the DL method exhibits a slight reduction in CNR compared to DAS, reflecting the inherent trade-off between improving resolution and preserving lumen-background contrast. Because the model operates pixel-wise, it cannot fully capture the spatial covariance structure that MVDR relies on, preventing it from completely reproducing MVDR's adaptive behavior. Performance is also sensitive to domain differences: models trained on phantom and carotid data may not generalize optimally to datasets with different probes, frequencies, or noise characteristics. Additionally, although CF is included as a feature, it is not used as a direct target, which may limit the model's ability to suppress clutter in highly incoherent regions.

Future extensions of this work could explore training on more diverse datasets, including additional in-vivo scans and multi-center acquisitions, to improve generalization. More expressive architectures—such as UNet-based RF-domain networks, beamformer-transformers, or covariance-aware neural networks—could enable direct learning from raw channel data and capture adaptive effects more comprehensively. Hybrid models that incorporate physical priorities, coherence cues, or differentiable beamforming operators may further enhance robustness. Finally, deploying the model on GPU or embedded ultrasound hardware, potentially via TensorRT or mobile inference engines, would enable real-time

adaptive beamforming suitable for portable and point-of-care imaging systems.

6. Conclusion

This work presented a deep-learning-based adaptive beamforming approach designed to approximate MVDR image quality while retaining the speed and simplicity of DAS. By training a lightweight multilayer perception on physically meaningful DAS-derived features and MVDR supervision, the model successfully learned pixel-wise intensity corrections that emulate key adaptive behaviors. Experiments on both phantom and in-vivo carotid datasets demonstrated that the proposed method improves lateral resolution and suppresses sidelobes compared to DAS, while producing image characteristics that closely resemble MVDR. Although a modest reduction in contrast was observed in some in-vivo cases, the model consistently enhanced structural clarity and provided more focused reconstructions. The results show that data-driven adaptive beamforming is a promising alternative to classical methods, offering a practical balance between computational efficiency and image quality. With further development—such as expanding training domains, incorporating RF-domain learning, or deploying real-time implementations—this approach may contribute to next-generation, high-performance ultrasound imaging systems.

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